

Socioeconomic Patterns in Diabetes Risk

Mini Data Science Project Proposal

1. Introduction

- Diabetes is influenced not only by biological factors but also by social and economic conditions such as income, education, and access to healthcare. This project will analyze a diabetes health indicators dataset that includes both medical and socioeconomic variables to see how these factors relate to diabetes status. The main goal is to understand which groups appear most vulnerable and how income, education, and access to care connect to diabetes risk.

2. Objectives

- Examine how diabetes prevalence changes across different IncomeBracket and EducationBracket values.
- Study the relationships between socioeconomic/access variables and health/lifestyle variables.
- Build a classification model that predicts whether a person has diabetes using mainly socioeconomic and access-to-care features, and compare it to a model that also includes lifestyle and clinical features.

3. Data Collection

- The project will use an existing diabetes health indicators dataset that is already available in CSV format. The data will be loaded using pandas, then cleaned and pre-processed before analysis.

4. Deliverables

- A Colab notebook that contains:
 - Visualizations such as bar charts of diabetes prevalence by income and education, boxplots or histograms of BMI and GeneralHealth across groups, and correlation/heatmap plots for key variables.
 - One or more classification models with performance metrics like accuracy, precision, recall, and confusion matrices, comparing the socioeconomic-only model to the full-features model.

5. Conclusion

By focusing on income, education, and access to healthcare, this project will explore the social side of diabetes risk using the provided dataset. The combination of visual analysis and predictive modeling should give a clearer picture of which groups are most at risk and how different factors interact, while also demonstrating practical skills in data cleaning, visualization, and machine learning.